

RESEARCH OVERVIEW & VISION

My research explores the **spatial, social, and design cognition of humans and AI** in built environments and social landscapes. Grounded in everyday, real-world experiences, my work draws on approaches from human-computer interaction (HCI) and **cognitive computational neuroscience** (cognitive science/psychology + neuroimaging + machine learning / deep learning).

With a long-term commitment to human well-being, my research advances two intertwined directions. First, I develop evidence-based computational design tools — integrating human experience data (neuroimaging, eye-tracking, and self-report) into AI-assisted workflows — to empower humans and AI to collaboratively shape architectural experiences throughout the design process ([C4], [C7]). Second, I examine how individuals represent and navigate their relationships with social issues — exploring the psychological and neural mechanisms of distance — and how conversational AI measures and reshapes these dynamics at scale.

In parallel, I pursue fundamental questions in ML/DL through a **NeuroAI** (brain-inspired / informed) lens, addressing shortcut learning ([C5]), biologically plausible propagation mechanisms, cognitive map formation in the hippocampus, and self-supervised learning grounded in cerebellar computation.

EDUCATION

National Tsing Hua University (NTHU), Hsinchu, Taiwan

03/2025 – Present

M.S. in Information Systems and Applications, Ranked 1st in Entering Cohort (expected direct-entry to Ph.D. program in Computer Science, Sep. 2026)

- Thesis topic: 'We Live in a Twilight World' — Establishing Cognitive-AI Benchmarking for Foundation Models through Bridging Human and Machine Intelligence in Architectural Design Process and Dwelling Experience.
- Course highlights: Music Information Retrieval; Computer Vision Pracice in Deep Learning; Data Science

National Yang Ming Chiao Tung University (NYCU), Taipei, Taiwan

09/2021 – 09/2024

M.S. in Cognitive Neuroscience, Ranked 1st in Entering Cohort (transferred)

- Thesis topic: Neural Encoding of Psychological Distance for Representing the Relationship between Individuals and Social Issues.
- Course highlights: HCI Research Methods; Advanced in 3D Computer Vision; ML; Statistics; Computational Corpus Linguistics

National Chiao Tung University (NCTU), Hsinchu, Taiwan

09/2014 – 07/2019

B.Sc. Biological Science & Technology; Minor in Design & Innovative Technology (Architecture/HCI)

- Thesis: Social Values Modulate Neural Correlates of Moral Decision Making.
- Fellowship for College Student Participation in Research Projects
- Neurobiology; Neuroscience Seminar; Cognitive Neuroscience; Functional Magnetic Resonance Imaging (fMRI); Sociology; Social studies of Science, Technology & Society (STS); Humanitarian Architecture, Architecture Theory; Design Fundamental; User-Centered Interaction Design

RESEARCH EXPERIENCE

Graduate Research Assistant, Human-Centered Machine Intelligence Lab, NTHU

02/2025 – Current

Supervisor: Po-Chih (Bruce) Kuo, Ph.D.

- Collected and curated a large-scale multimodal house tour video dataset to train/fine-tune a foundation model.
- Conducted and wrote two CHI papers on a human-informed, AI-native architectural design tool ([C4], [C7]).

Research Intern, AI4SG Lab, Dept. of CS, National University of Singapore

06/2023 – 09/2025

Supervisor: Yi-Chieh (EJ) Lee, Ph.D.

- Provided theoretical grounding in Construal Level Theory (CLT), study design, and statistical analysis across two studies on psychological distance and information-seeking behavior in conversational AI, showing preference for AI in abstract/future-oriented tasks and traditional search for concrete/immediate needs ([C3]), and improved trust/satisfaction via distance-aligned response formats ([C6]).

Research Assistant, Human-Centered Machine Intelligence Lab, NTHU

10/2020 – 02/2025

Supervisor: Po-Chih (Bruce) Kuo, Ph.D.

- Played a leading role in developing an interdisciplinary research proposal investigating human architectural experience in interior environments through neuroimaging, eye-tracking, and deep learning. The project received NT\$4.73 million in funding from the National Science and Technology Council, Taiwan.
- Led an industry-collaborative research project applying human computation methods to the evaluation of olfactory computing interfaces, resulting in multiple peer-reviewed publications ([J1], [C1], [C2]).

Research Assistant, Cognitive Neuroimaging Lab, NYCU

03/2021 – 08/2021

Supervisor: Nissen Wen-Jui Kuo, Ph.D.

- Contributed to the implementation of inter-subject correlation (ISC) and representational similarity analysis (RSA) pipelines for naturalistic fMRI data analysis, examining neural response patterns across signers and sign language learners ([P3]).
- Conducted a comprehensive literature review on cerebellar cognition and advanced neuroimaging methods in cerebellar neuroimaging.

Undergraduate Research Assistant, Brain and Cognitive Sciences Lab, NCTU

06/2016 – 06/2019

Advisor: Chih-Mao Huang, Ph.D.

- Recruited and scanned 50+ participants across countries, and conducted fMRI analyses examining how socio-cultural values modulate neural correlates of moral evaluation, culminating in a full journal manuscript draft ([P1], [P2], [WIP1]).

PUBLICATIONS

JOURNAL ARTICLES

[J1] Lin, Y.C., Yu, S. L., Zhuang, A. Y., Lee, C., ... & Kuo, P. C. (2024). Representing Scents: An Evaluation Framework of Scent-related Experiences. International Journal of Human-Computer Studies, 103357.

CONFERENCE PAPERS

- [C7] Lin, Y.C., Chien, C.-Y., Hou, K.-H., Chen, H.-Y., & Kuo, P.-C. (2026). AIDED: Augmenting Interior Design with Human Experience Data for Designer–AI Co-Design. CHI'26, Barcelona. (25.3% acceptance)
- [C6] Yang, Y., Tan, Y., King, J. T., Lin, Y.C. & Lee, Y. C. (2026). Fit Matters: Format-Distance Alignment Improves Conversational Search. CHI'26, Barcelona. (25.3% acceptance)
- [C5] Lin, Y.C., Lee, C.Y., & Kuo, P.-C. (2026). Play the (Mis)Match: fMRI-Aligned Feature Fine-Tuning to Reveal Shortcut Bias in DNNs. AAAI 2026 NeuroAI Workshop, Singapore. (10/77 accepted)
- [C4] Lin, Y.C., Chung, W.-Y., Chien, C.-Y., Su, C.-H., & Kuo, P.C. (2025). Shaping Architectural Design Tools Through HCI. CHI'25 EA, Yokohama. (32.8% acceptance)
- [C3] Yang, Y., Tan, Y., Lin, Y.C., King, J., Liu, Z., & Lee, Y. (2025). Psychological Distance in Conversational vs. Web Search. CHI'25, Yokohama. (24.9% acceptance)
- [C2] Peng, H.-P., Hsiao, H.-L., Su, C.-H., Lin, Y.C. & Kuo, P. C. (2024). Deciphering Odor Perception Through EEG and Gas Sensors. IEEE EMBC 2024, Orlando.
- [C1] Zhuang, A. Y., Lin, Y.C., Yu, S. L., & Kuo, P. C. (2021). A Brain-Sensing Fragrance Diffuser for Mental State Regulation. IEEE ICHMS 2021, pp. 1–6.

POSTER PRESENTATIONS

- [P4] Lin, Y.C., Chung, W.Y., & Kuo, P.-C. (2025). Distinct Functional Connectivity Attenuations in Residential Interior Styles. ANFA 2025, Salk Institute, San Diego.
- [P3] Lin, Y.C., Chiu, Y. S., & Kuo, W. J. (2021). Neural Basis of Sign Language Proficiency via Naturalistic fMRI and ISC. SfN 2021, San Diego.
- [P2] Lin, Y.C. & Huang, C. M. (2020). Socio-cultural Differences in Neural Underpinnings of Perceiving Built Environments. ANFA 2020 (Virtual).
- [P1] Lin, Y.C., Doole, R., Huang, H. W., & Huang, C. M. (2019). Social Values Modulate Neural Correlates of Moral Decision Making. CNS 2019, San Francisco.

WORK IN PROGRESS

- [WIP1] Lin, Y.C., Li, Y. S., Huang, H. W., Fan, Y. T., Doole, R., Wu C. W., Tzeng, Ovid J. L., Cheng, Y., & Huang, C. M. Cross-Cultural Social Values Modulate Neural Correlates of Moral Decision-Making. (under review)

RESEARCH SKILLS

- Development:** Development: Python/C++; Git; Docker; Claude Code; Figma; Blender; full-stack web development (React, Node.js, MongoDB)
- Deep Learning:** representation learning; self-supervised learning; brain–model alignment; computer vision (3D/video); large language model fine-tuning
- Neuroimaging:** fMRI/EEG/eye-tracking experiment design & analysis; naturalistic paradigm analysis (ISC, RSA, encoding model); motion capture; physiological signal recording & analysis
- Empirical approach :** behavioral experiment; psychophysics; formative study; user interview; qualitative analysis; statistics

ACADEMIC SERVICE

- Reviewer: CHI'23 (LBW), CHI'25 (LBW), CHI'26, IUI'26, CogSci'26, Nature Scientific Data
- Student Volunteer: TAICHI 2020 Conference, Taipei, Taiwan

SELECTED FUNDING & AWARDS

- 2025, 2026** Student Travel Grant, National Science and Technology Council (NSTC), Taiwan
- 2025** PhD Research Fellowship, National Science and Technology Council (NSTC), Taiwan
- 2018** Student Travel Grant, Higher Education Sprout Project, Ministry of Education, Taiwan
- 2017** Fellowship for College Student Participation in Research Projects, MoST, Taiwan

REFERENCES

Dr. Po-Chih (Bruce) Kuo Associate Professor, Dept. of CS, National Tsing Hua University (NTHU), Taiwan <i>Research Assistant Supervisor</i>	Dr. Chih-Mao Huang Associate Professor, Dept. of Psychology, The University of Hong Kong (HKU), Hong Kong <i>Bachelor's Thesis Supervisor</i>	Dr. Yi-Chieh (EJ) Lee Assistant Professor, AI4SG Lab, Dept. of CS, National University of Singapore (NUS) <i>Research Intern Supervisor</i>
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